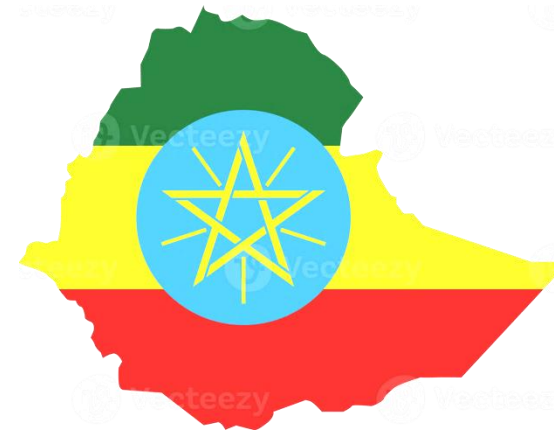




MAED-Based Analysis of Rural Household Cooking Energy Demand in Ethiopia



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Outline

- ✓ Context, challenges & Research Question
- ✓ Ethiopian Reference Energy System
- ✓ Scenarios
- ✓ Results
- ✓ Conclusions, Policy insights, & Future Work

Context

- Rural Ethiopia heavily depends on traditional biomass fuels for cooking, used by over 95% of households.
- This reliance drives deforestation, environmental degradation, and serious health risks, especially for women and children.
- Despite clean energy commitments, household cooking energy planning lacks data-driven, long-term projections.
- The IAEA's MAED model offers a systematic, scenario-based approach to estimating energy demand.
- Applied to rural cooking, it supports evidence-based policy decisions on improved cookstoves and clean fuel adoption to reduce health and environmental impacts.

Challenges and Research Question

❑ Main challenges investigated

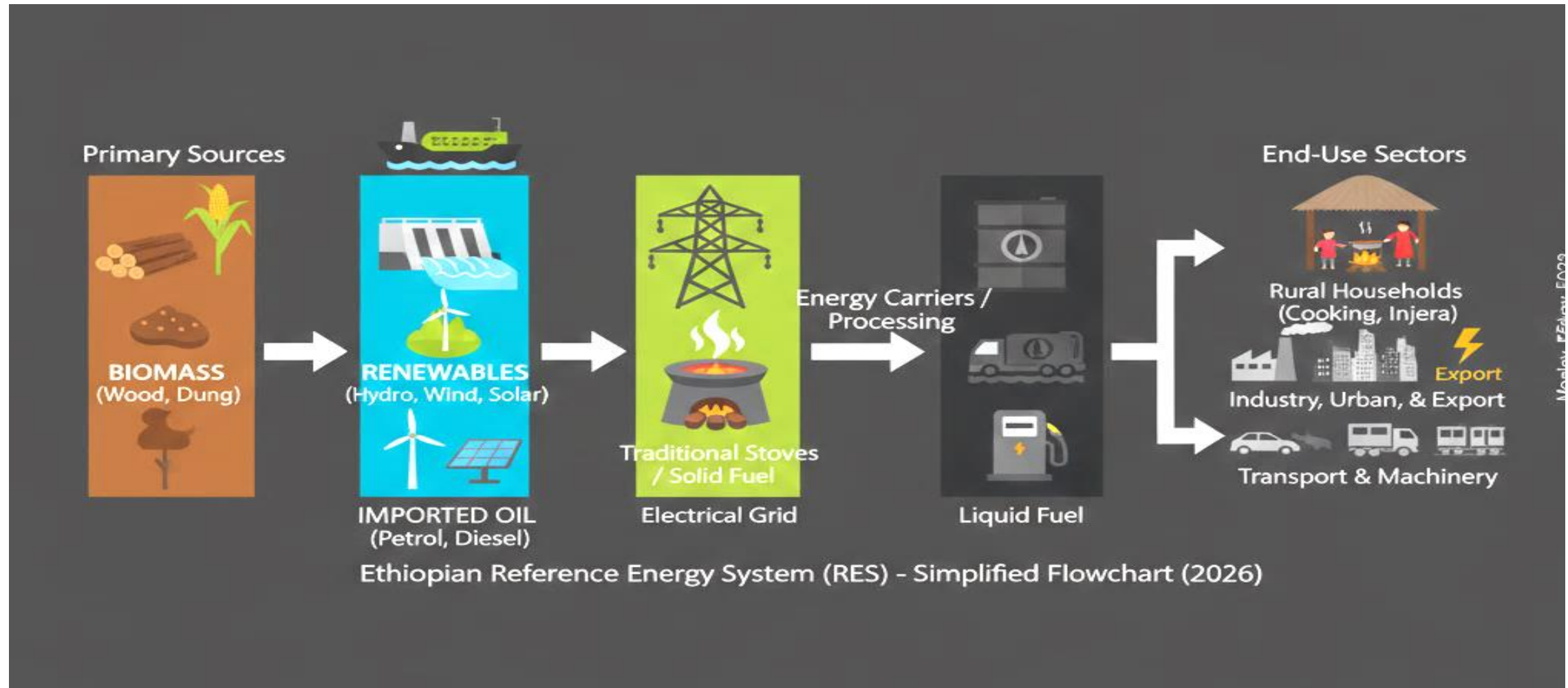
The continued and increasing dependence of rural households in Ethiopia on traditional biomass fuels for cooking is driven by rapid population growth and the low adoption of efficient and clean cooking technologies

❑ Research question

To what extent can improved cooking technologies and fuel switching reduce traditional biomass dependence in rural Ethiopia?



Ethiopian Reference Energy System

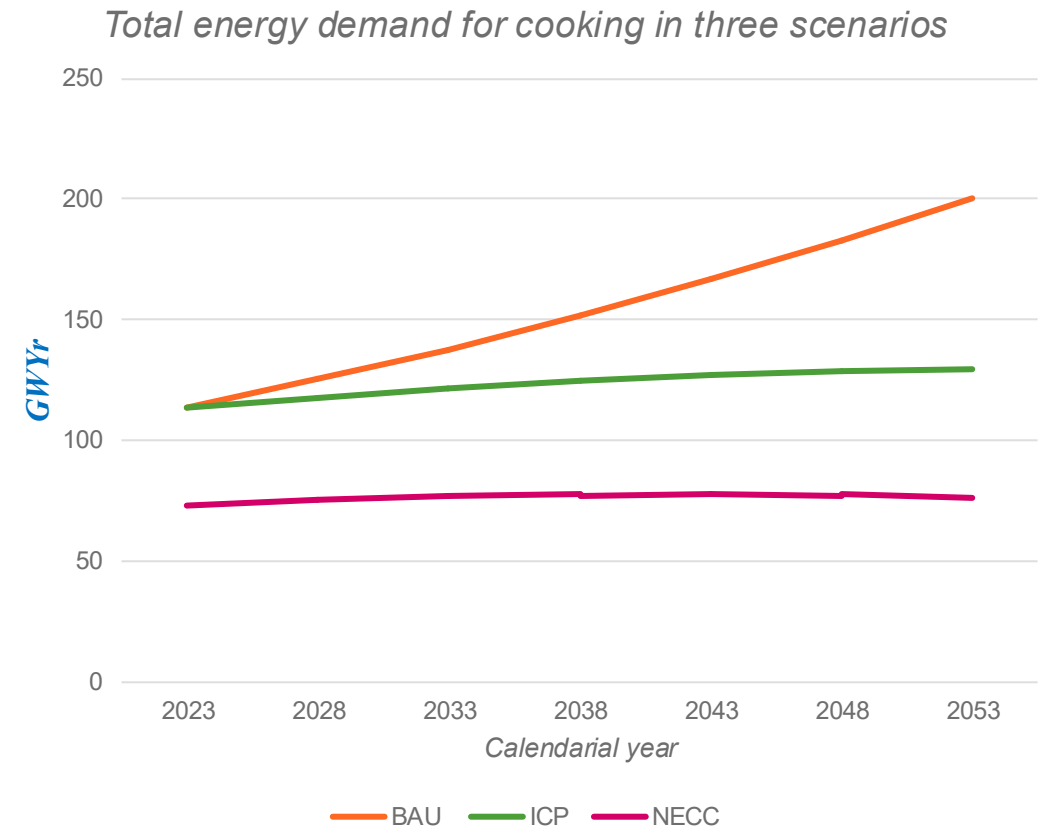


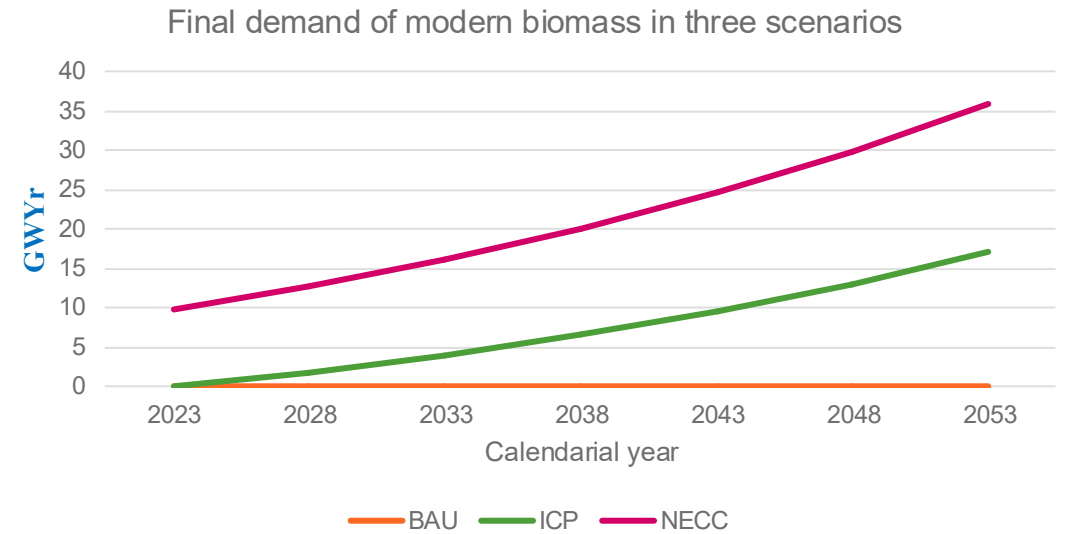
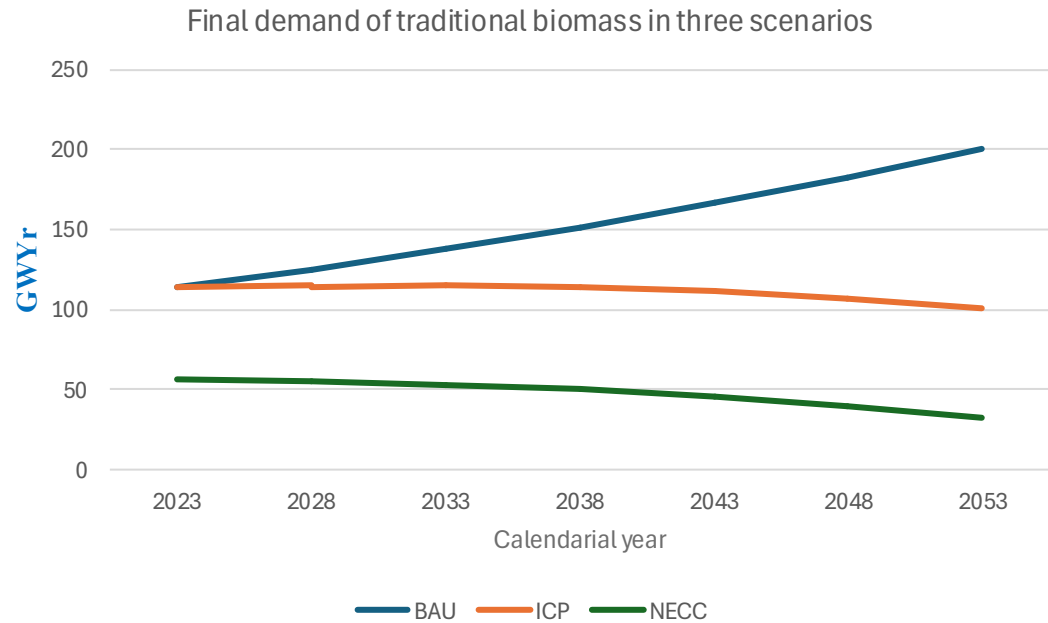
Scenarios

SCENARIO LABEL	SCENARIO DESCRIPTION	KEY ASSUMPTION
BAU (business as usual)	Reflects current trends in the rural cooking energy sector without new policy interventions or technological disruptions	✓ No major change in cooking technology; continued reliance on biomass; ✓ Steady population growth; ✓ Health and environmental impacts remain similar to current trends.
Improved cookstove penetration	Accelerated adoption of improved biomass stoves and partial fuel switching to electricity, driven by rural energy access programs	✓ Reduce demand of traditional biomass by 50% by 2053; ✓ Growth rate of GDP ✓ Technology adoption
National electrified and clean cooking	High-ambition transition to advanced electric enabled by national electrification and strong policies	✓ High penetration of electricity for cooking by 2053, ✓ Widespread adoption of clean cooking technology, like electric stove or and environmental benefit

Results and Discussion

Scenario label	Final energy demand GWyr	Demand reduction (%)
BAU (business as usual)	200.60	0
ICP (Improved cookstove penetration)	129.25	35.56
NECC (National electrified and clean cooking)	75.88	62.4





Parameter	Initial State	Final State	Reduction / Change	Percentage of reduction
Energy Demand	200.23 GWyr	75.00 GWyr	125.23 GWYr	62.4
Carbon Intensity	8.0 MtCO2/GWyr	4.0 MtCO2/GWyr	4.0 MtCO2/GWYr	50
Total Emissions	1,601.84 MtCO2	300.00 MtCO2	1,301.84 MtCO2	81.28

Key Policy Insights & Government Strategy

- **Targeting Tier 3+ Transition:** Ethiopia aims to elevate at least 50% of all cooking appliances to Tier 3 or above (highly efficient/clean) by 2035.
- **Integrated Electrification:** The National Electrification Program (NEP 2.0) and the National Clean Cooking Roadmap (2025–2035) are being integrated to leverage grid and off-grid expansion for e-cooking (electric cooking).
- **Decoupling Biomass from Growth:** Modeling shows that without intervention, traditional biomass demand will nearly double by 2050.
- **Gender and Health Empowerment:** A central pillar is the empowerment of 2 million women by 2035 through training and leadership in the clean cooking sector, reducing the 64,000 annual premature deaths attributed to indoor air pollution.

Policy recommendations to Ethiopian government (Ministry of Energy)

1. Operationalize a "Tier-3 Minimum" Standard for rural markets
2. Launch a "National e-Cooking Subsidy" linked to NEP 2.0
3. Formalize a commercial "Modern Biomass" value chain
4. Leverage climate finance (Carbon Credits) for rural subsidies

Conclusions

Compounding Environmental Gains

The total CO₂ reduction is compounded, Because using less energy and (presumably) a cleaner fuel, the carbon savings are exponential rather than linear.

- ✓ The most significant takeaway is that the scenario demonstrates a deep structural shift. Reducing energy demand by over 62.4% through a fuel switch

THANK YOU FOR YOUR
ATTENTION!!